KUN PANG (HENDRIX)

Ph.D. Student, Electrical & Computer Engineering · University of Virginia

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Education

University of Virginia — Charlottesville, VA, USA
Ph.D. in Electrical & Computer Engineering — Aug 2026 – Present
Advisor: Prof. Steven Bowers · Research focus: RFIC design

Sun Yat-sen University — Guangzhou, China
B.Eng. in Information Engineering (Elite Class) — 2022 – 2026
GPA: 3.7/4.0 · Core courses (DSP, Communication Theory, Radar): 4.0/4.0 · TOEFL: 104

Research Experience

K-Band Reconfigurable Intelligent Surface (RIS) for Millimeter-Wave Systems — Dec 2024 – Dec 2025
Prof. Shaoqiu Xiao, Sun Yat-sen University

- Designed a multilayer K-band RIS unit cell (20–26 GHz) featuring two rotationally symmetric hollow I-shaped resonant layers on RO3003 substrate, optimized via HFSS with Floquet-port simulation; unit size 4×4 mm ($\approx 0.30\lambda$ at 22 GHz)
- Achieved $\sim 360°$ phase tuning range with $\sim 120°$ step per coding state, reflection loss <1.6 dB for four primary states, and excellent phase linearity ($R^2 = 0.9991$, RMSE $= 0.515°$) within the 22–23 GHz operating band
- First-author paper published at EuCAP 2026, Dublin (see Publications)

Microstrip Patch Antenna Array Design — Mar 2024 – Dec 2024
Prof. Shaoqiu Xiao, Sun Yat-sen University

- Designed and fabricated a 4-element patch array at 5.8 GHz (isolation < -20 dB); characterized prototype via VNA measurements and anechoic chamber radiation testing

Engineering Projects

Multilayer Dual Circularly Polarized Satellite Antenna (13–14.5 GHz) — Oct 2025 – Mar 2026
Prof. Shaoqiu Xiao, Sun Yat-sen University

- Designed a Ku-band dual-CP multilayer satellite antenna with isolation-enhanced structure; performed full-cycle HFSS optimization, simulation, and prototype testing

UAV Communication Link Modelling — Dec 2025
Prof. Xiang Chen, Sun Yat-sen University

- Built a full end-to-end link simulator in MATLAB (QPSK, Turbo coding at rates 1/3 and 2/3, CRC-16, pilot-based equalization) to validate a proposed UAV frame structure
- Modelled a 3-path multipath fading channel with delay spread up to 400 ns and max Doppler 737 Hz; confirmed frame integrity via BER/BLER statistics

BPSK Digital Voice Transmission System — May 2025
Prof. Xiang Chen · Lingte Cup National SDR & Communications Design Competition

- Built a real-time end-to-end digital voice link using PlutoSDR and GNU Radio: channel coding, BPSK modulation, matched filtering, and equalization; BER $< 10^{-6}$

Satellite Communication Terminal Support — Oct 2025
Prof. Xiang Chen, Industry Collaboration

On-site satellite modem link setup and RF testing; one of the few undergraduates working alongside graduate and doctoral researchers in an industry field environment.

Publications

1. **K. Pang**, Y. Chen, H. Lu, B. Lei, and S. Xiao, "Enhanced Linear Phase Control of a K-Band Reconfigurable Intelligent Surface via Coupled Hollow I-Shaped Resonators," *Proc. EuCAP 2026*, Dublin, Ireland, pp. 1–5, 2026. DOI: 10.23919/EuCAP68105.2026.11612526
2. B. Lei, Y. Li, G. Xu, C. Gao, **K. Pang**, Y. Fu, Z. Yu, Y. Shi, and S. Xiao, "Radiation performance analysis of asymmetric-structured self-biased magnetoelectric antennas via multi-field coupling modeling," *Smart Materials and Structures*, 2025. DOI: 10.1088/1361-665X/ae1beb
3. B. Lei, Y. Li, G. Xu, C. Gao, **K. Pang**, D. Ye, Y. Shi, and S. Xiao, "Theoretical analysis on radiation performance of magnetoelectric antennas considering eddy-current loss and quasistatic hysteresis effect," *Journal of Applied Physics*, vol. 138, no. 4, p. 044101, 2025. DOI: 10.1063/5.0282234

Awards & Honors

- First Prize, Lingte Cup National SDR & Communications Design Competition — 2025

- Outstanding Student Leader Award, Sun Yat-sen University 2026
- Excellent Student Scholarship, Sun Yat-sen University 2023, 2025

Selected Coursework

Electromagnetic Fields and Waves (4.0/4.0) · Digital Signal Processing (4.0/4.0) · Communication Theory (4.0/4.0) · Automatic Control Theory (4.0/4.0)

Technical Skills

- **RF & EM Simulation:** HFSS, ADS
- **RF Lab & Measurement:** VNA, Anechoic chamber
- **SDR & Communications:** GNU Radio, PlutoSDR, BPSK/QPSK/QAM, channel coding
- **Programming:** MATLAB/Simulink, Python, C++, Git

Languages

English — Proficient (TOEFL 104) · Chinese (Mandarin) — Native